

Global Data Jobs Market Analysis using Python

Project Objective

This project analyses global data-related job postings to identify patterns in salary trends, remote work opportunities, hiring demand, company-based salary differences, and country-level job market behaviour using Python.

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datasets import load_dataset
```

```
In [2]: df=pd.read_csv('/content/data_jobs.csv', engine='python', on_bad_lines='skip')
```

```
In [3]: df.head()
```

Out[3]:

	job_title_short	job_title	job_location	job_via	job_schedule_type
--	-----------------	-----------	--------------	---------	-------------------

0	Senior Data Engineer	Senior Clinical Data Engineer / Principal Clin...	Watertown, CT	via Work Nearby	Full-time
---	----------------------	---	---------------	-----------------	-----------

1	Data Analyst	Data Analyst	Guadalajara, Jalisco, Mexico	via BeBee México	Full-time
---	--------------	--------------	------------------------------	------------------	-----------

2	Data Engineer	Data Engineer/Scientist/Analyst, Mid or Senior...	Berlin, Germany	via LinkedIn	Full-time
---	---------------	---	-----------------	--------------	-----------

3	Data Engineer	LEAD ENGINEER - PRINCIPAL ANALYST - PRINCIPAL ...	San Antonio, TX	via Diversity.com	Full-time
---	---------------	---	-----------------	-------------------	-----------

4	Data Engineer	Data Engineer- Sr Jobs	Washington, DC	via Clearance Jobs	Full-time
---	---------------	------------------------	----------------	--------------------	-----------



```
In [4]: df.tail()
```

Out[4]:

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_w
785736	Software Engineer	DevOps Engineer	Singapura	melalui Trabajo.org	Pekerjaan tetap	
785737	Data Analyst	CRM Data Analyst	Bad Rodach, Jerman	melalui BeBee Deutschland	Pekerjaan tetap	
785738	Business Analyst	Commercial Analyst - Start Now	Malaysia	melalui Ricebowl	Pekerjaan tetap	
785739	Data Engineer	Principal Associate, Data Engineer (Remote-Eli...	Newark, New Jersey, Amerika Serikat	melalui Recruit.net	Pekerjaan tetap	
785740	Software Engineer	AWS System Analyst	India	melalui Trigyn	Pekerjaan tetap	

In [5]: `df.sample()`

Out[5]:

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_fr
633229	Data Engineer	Data Engineer	Leicester, UK	via Totaljobs	Full-time	

In [6]: `df.shape`

Out[6]: (785741, 17)

In [7]: `df.columns`

Out[7]: Index(['job_title_short', 'job_title', 'job_location', 'job_via', 'job_schedule_type', 'job_work_from_home', 'search_location', 'job_posted_date', 'job_no_degree_mention', 'job_health_insurance', 'job_country', 'salary_rate', 'salary_year_avg', 'salary_hour_avg', 'company_name', 'job_skills', 'job_type_skills'], dtype='object')

In [8]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 785741 entries, 0 to 785740
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   job_title_short                       785741 non-null  object
1   job_title                             785740 non-null  object
2   job_location                         784696 non-null  object
3   job_via                               785733 non-null  object
4   job_schedule_type                    773074 non-null  object
5   job_work_from_home                   785741 non-null  bool
6   search_location                      785741 non-null  object
7   job_posted_date                      785741 non-null  object
8   job_no_degree_mention                785741 non-null  bool
9   job_health_insurance                 785741 non-null  bool
10  job_country                          785692 non-null  object
11  salary_rate                          33067 non-null   object
12  salary_year_avg                      22003 non-null   float64
13  salary_hour_avg                     10662 non-null   float64
14  company_name                         785723 non-null  object
15  job_skills                           668704 non-null  object
16  job_type_skills                      668704 non-null  object
dtypes: bool(3), float64(2), object(12)
memory usage: 86.2+ MB
```

In [9]: `df.describe()`

Out[9]:

	salary_year_avg	salary_hour_avg
count	22003.000000	10662.000000
mean	123286.274072	47.016598
std	48312.449482	21.890738
min	15000.000000	8.000000
25%	90000.000000	27.500000
50%	115000.000000	45.980000
75%	150000.000000	61.159996
max	960000.000000	391.000000

Data Loading & Initial Exploration

The dataset was loaded using Pandas to analyse global data-related job postings. Initial exploration was performed using functions such as `head()`, `info()`, and `describe()` to understand the dataset structure, available columns, data types, and salary information.

This step helped identify:

- salary-related columns
- remote job information
- company details
- country-based job distribution
- missing values in salary columns

In [10]: `df.isnull().sum()`

Out[10]:

	0
job_title_short	0
job_title	1
job_location	1045
job_via	8
job_schedule_type	12667
job_work_from_home	0
search_location	0
job_posted_date	0
job_no_degree_mention	0
job_health_insurance	0
job_country	49
salary_rate	752674
salary_year_avg	763738
salary_hour_avg	775079
company_name	18
job_skills	117037
job_type_skills	117037

dtype: int64

In [11]: `df.duplicated().sum()`

Out[11]: `np.int64(101)`

In [12]: `df.drop_duplicates()`

Out[12]:

	job_title_short	job_title	job_location	job_via	job_schedu
0	Senior Data Engineer	Senior Clinical Data Engineer / Principal Clin...	Watertown, CT	via Work Nearby	Fi
1	Data Analyst	Data Analyst	Guadalajara, Jalisco, Mexico	via BeBee México	Fi
2	Data Engineer	Data Engineer/Scientist/Analyst, Mid or Senior...	Berlin, Germany	via LinkedIn	Fi
3	Data Engineer	LEAD ENGINEER - PRINCIPAL ANALYST - PRINCIPAL ...	San Antonio, TX	via Diversity.com	Fi
4	Data Engineer	Data Engineer- Sr Jobs	Washington, DC	via Clearance Jobs	Fi
...	...	...	...	...	...
785736	Software Engineer	DevOps Engineer	Singapura	melalui Trabajo.org	Pekerjaa
785737	Data Analyst	CRM Data Analyst	Bad Rodach, Jerman	melalui BeBee Deutschland	Pekerjaa
785738	Business Analyst	Commercial Analyst - Start Now	Malaysia	melalui Ricebowl	Pekerjaa
785739	Data Engineer	Principal Associate, Data Engineer (Remote-Eli...	Newark, New Jersey, Amerika Serikat	melalui Recruit.net	Pekerjaa
785740	Software Engineer	AWS System Analyst	India	melalui Trigyn	Pekerjaa

785640 rows × 17 columns



```
In [13]: df['job_posted_date']=pd.to_datetime(df['job_posted_date'])
```

```
In [14]: salary_df = df[df['salary_year_avg'].notnull()]
```

```
In [15]: df['salary_hour_avg']=df['salary_hour_avg'].fillna(df['salary_hour_avg'].mean())
```

Data Cleaning & Preparation

Data cleaning techniques were applied to improve data quality before analysis.

The following methods were used:

Handling missing salary values using notnull() Removing duplicate records Converting date columns using to_datetime() Filtering valid salary records for accurate salary analysis Checking data types using dtypes

Instead of replacing salary null values with averages, salary analysis was performed only on rows containing valid salary data. This improved the realism and reliability of salary-based insights.

1.Which top 10 job titles have the highest average salaries?

```
In [16]: Highest_salary=df.groupby('job_title_short')['salary_year_avg'].mean().sort_value  
Highest_salary.head(10).reset_index()
```

```
Out[16]:
```

	job_title_short	salary_year_avg
0	Senior Data Scientist	154206.292996
1	Senior Data Engineer	145840.611624
2	Data Scientist	135988.837171
3	Data Engineer	130125.604250
4	Machine Learning Engineer	126774.315972
5	Senior Data Analyst	113911.363665
6	Software Engineer	113393.760054
7	Cloud Engineer	111268.453846
8	Data Analyst	93841.907854
9	Business Analyst	91082.612833

Logic Used

The dataset was grouped using the job_title_short column to compare average salaries across different job roles.

The groupby() function was used to combine similar job titles, while the mean() function calculated the average yearly salary for each role using the salary_year_avg column. The results were then sorted in descending order using sort_values(ascending=False) to rank the highest-paying job titles from highest to lowest salary. Finally, the head(10) function was used to extract the top 10 highest-paying job roles, and reset_index() was applied to convert the grouped index into a clean tabular format for easier readability and visualisation.

This analysis helped identify the most highly paid job positions within the global data analytics industry.

Methods & Techniques Used Data Grouping, Aggregation Analysis, Salary Analysis, Ranking Analysis, Sorting Techniques, Top-N Analysis, Exploratory Data Analysis (EDA).

Functions Used `groupby()`, `mean()`, `sort_values()`, `head()`, `reset_index()`.

Analysis

The analysis revealed that Senior Data Scientist roles have the highest average salary at approximately 154K, followed by Senior Data Engineer roles at around 145K. Regular Data Scientist and Data Engineer positions also showed strong salary performance above \$130K.

This indicates that:

seniority significantly increases salary potential technical AI and engineering roles dominate the highest-paying positions companies highly value advanced analytical and engineering expertise

2. Which country has the highest number of remote jobs?

```
In [17]: Higits_remote_job=df.groupby('job_country')['job_work_from_home'].sum().sort_val
Higits_remote_job.head(10).reset_index()
```

```
Out[17]:
```

	job_country	job_work_from_home
0	United States	23021
1	India	5712
2	Sudan	3930
3	United Kingdom	3764
4	Canada	2812
5	Spain	2645
6	Poland	1857
7	France	1694
8	Mexico	1619
9	Brazil	1542

Logic Used

The dataset was grouped using the job_country column to compare the number of remote jobs available in different countries.

The groupby() function was used to combine job records by country, while the sum() function was applied on the job_work_from_home column to calculate the total number of remote job opportunities in each country.

The results were then sorted in descending order using sort_values(ascending=False) to identify countries with the highest remote job availability.

Finally, the head(10) function was used to extract the top 10 countries with the largest number of remote jobs, and reset_index() converted the grouped output into a cleaner tabular structure for analysis and visualisation.

This analysis helped identify which countries currently provide the strongest remote work opportunities in the global data job market.

Methods & Techniques Used Data Grouping, Aggregation Analysis, Remote Work Analysis, Ranking Analysis, Sorting Techniques, Top-N Analysis, Exploratory Data Analysis (EDA).

Functions Used groupby(), sum(), sort_values(), head(), reset_index().

Analysis

The United States recorded the highest number of remote jobs with more than 23,000 remote positions, followed by India with around 5,700 jobs and the United Kingdom with approximately 3,700 remote opportunities.

This suggests that:

the US leads the global remote hiring market remote work adoption is strongest in developed technology economies India is becoming a major global remote workforce hub

3.What are the top-paying entry-level roles?

```
In [18]: Entry_level_job=df[df['job_title'].str.contains('Entry Level',case=False,na=False)]
top_paying_entry_level_roles=Entry_level_job.groupby('job_title')['salary_year_avg'].head(10).reset_index()
```


Out[18]:

	job_title	salary_year_avg
0	Entry Level PhD Data Scientist (Boston, MA)	132500.0
1	Entry Level Data Scientist	112500.0
2	Entry Level - Associate Data Scientist Event - ...	107000.0
3	Entry Level - Associate Data Scientist - Comme...	107000.0
4	Data Scientist – Entry Level 2024	102100.0
5	Data Scientist (Entry Level)	100000.0
6	Entry Level Programmer/Data Scientist/Analyst/...	98000.0
7	ENTRY LEVEL DATA SCIENTIST	97500.0
8	entry level data scientist	97500.0
9	Entry Level - Associate Data Scientist (Federal)	92500.0

Logic Used

The dataset was filtered using the job_title column to identify entry-level job roles. The str.contains() function was applied with the keyword "Entry Level" to extract only beginner-level positions from the dataset.

After filtering the relevant records, the dataset was grouped using the job_title column to compare salaries across different entry-level roles.

The mean() function was then used to calculate the average yearly salary for each entry-level position using the salary_year_avg column.

The results were sorted in descending order using sort_values(ascending=False) to rank the highest-paying entry-level roles from highest to lowest salary.

Finally, the head(10) function extracted the top 10 highest-paying entry-level positions, while reset_index() converted the grouped output into a structured tabular format for better readability and visualisation.

This analysis helped identify which beginner-level data roles currently offer the strongest salary opportunities in the job market.

Methods & Techniques Used Data Filtering, Text Filtering, Conditional Filtering, Data Grouping, Aggregation Analysis, Salary Analysis, Ranking Analysis, Sorting Techniques, Top-N Analysis, Exploratory Data Analysis (EDA).

Functions Used str.contains(), groupby(), mean(), sort_values(), head(), reset_index().

Analysis

The highest-paying entry-level role identified was “Entry Level PhD Data Scientist” with an average salary of approximately 132K. StandardEntryLevelDataScientistpositionsalsoshowedstrongsalariesabove 112K.

This indicates:

Data Science offers excellent salary opportunities even at beginner level advanced technical and research-based entry roles receive premium compensation AI-focused careers remain highly valuable in the job market

4.Question: Which Job Titles Have the Lowest and Highest Average Salaries?

```
In [21]: df.groupby('job_title')['salary_year_avg'].mean().sort_values().reset_index()
```

Out[21]:

	job_title	salary_year_avg
0	Data Engineer - Hadoop	15000.0
1	Showwcase Elite - Data Engineer	20000.0
2	Data Engineer Jobs In Flipdish for Engineering...	24000.0
3	Data Engineer (ETL) (TS/SCI + Poly)	25000.0
4	#DISCOVER I 2024 Improvement based in Data Ana...	25920.0
...	...	...
234668	 R&D Engineer	NaN
234669	 Project Lead	NaN
234670	 Senior Data Engineer	NaN
234671	 BI Data Engineer - Medtech - H/F	NaN
234672	 Data Engineer - Medtech - H/F	NaN

234673 rows × 2 columns

Logic Used

The dataset was grouped using the job_title column to compare average salaries across all job roles.

The groupby() function was used to combine similar job titles together, while the mean() function calculated the average yearly salary for each role using the salary_year_avg column.

The results were then sorted using `sort_values()` to arrange job roles based on their average salary values. This helped identify both lower-paying and higher-paying positions within the dataset.

Finally, `reset_index()` was applied to convert the grouped index into a clean tabular format for easier readability and further analysis.

This analysis helped compare salary distribution across different data-related job positions and identify salary differences between various roles in the industry.

Methods & Techniques Used Data Grouping, Aggregation Analysis, Salary Analysis, Ranking Analysis, Sorting Techniques, Exploratory Data Analysis (EDA). Functions Used `groupby()`, `mean()`, `sort_values()`, `reset_index()`.

Analysis

The salary comparison revealed that some lower-paying Data Engineering roles earned salaries around $15K-25K$, *whileseniorandhighlyspecializedpositionsexceeded* 300K+ salaries.

This shows:

salary differences in the data industry are extremely large specialization and seniority heavily impact earnings advanced technical roles generate significantly higher compensation

5.Which Data Analyst Roles Earn the Highest Average Salaries?

```
In [22]: Data_analyst_jobs=df[df['job_title_short']=='Data Analyst']
Data_analyst_jobs.groupby('job_title')['salary_year_avg'].mean().sort_values(asc
```

Out[22]:

	salary_year_avg
job_title	
Data base administrator	400000.0
Head of Infrastructure Management & Data Analytics - Financial...	375000.0
Director of Safety Data Analysis	375000.0
HC Data Analyst , Senior	375000.0
Director of Analytics	336500.0
...	...
🤖 Data Analyst Assistant (Fintech, Python, A lvl / Diploma ok)	NaN
🚀 DATA ARCHITECT	NaN
💛 Data Strategy Consultant	NaN
💛 Junior Data Strategy Consultant	NaN
🤖 AWS Data Analyst 🤖 🌱 Custom software solutions company from...	NaN

60957 rows × 1 columns

dtype: float64

Logic Used

The dataset was first filtered using the job_title_short column to include only Data Analyst-related job roles.

Conditional filtering was applied to extract records where the job category was "Data Analyst".

After filtering the dataset, the groupby() function was used on the job_title column to compare salaries across different Data Analyst positions and specialisations.

The mean() function calculated the average yearly salary for each role using the salary_year_avg column.

The results were then sorted in descending order using sort_values(ascending=False) to identify which Data Analyst roles offer the highest average salaries.

This analysis helped compare salary differences between various Data Analyst positions and identify higher-paying analyst specialisations in the job market.

Methods & Techniques Used Data Filtering, Conditional Filtering, Data Grouping, Aggregation Analysis, Salary Analysis, Ranking Analysis, Sorting Techniques, Exploratory Data Analysis (EDA). Functions Used groupby(), mean(), sort_values().

Analysis

The highest-paying analyst-related role identified was Database Administrator with an average salary of approximately

400K, followed by leadership and senior healthcare analytics roles earning around 375K.

This suggests:

specialized analytical roles command premium salaries leadership and healthcare-related analytics positions are highly valued advanced analytical expertise strongly improves compensation

6.What Percentage of Jobs Are Fully Remote?

```
In [23]: Job_from_Home=df[df['job_work_from_home']==True]
remote_job_counts=Job_from_Home['job_title'].value_counts()

total_remote_jobs=len(Job_from_Home)

remote_job_percentages=(remote_job_counts/total_remote_jobs)*100
display(remote_job_percentages.head(10).reset_index())
```

	job_title	count
0	Data Engineer	8.218311
1	Data Scientist	4.528986
2	Senior Data Engineer	4.222740
3	Data Analyst	3.779906
4	Senior Data Scientist	1.966874
5	Senior Data Analyst	0.983437
6	Azure Data Engineer	0.625431
7	Lead Data Engineer	0.613929
8	Machine Learning Engineer	0.612491
9	Lead Data Scientist	0.460087

Logic Used

The dataset was filtered using the job_work_from_home column to extract only fully remote job postings where the value was True.

After filtering the dataset, the value_counts() function was applied on the job_title column to calculate the number of remote job postings available for each job role.

The total number of remote jobs was then calculated using the `len()` function.

Next, percentage analysis was performed by dividing the number of remote jobs for each role by the total number of remote jobs and multiplying the result by 100.

Finally, the results were displayed using `head(10)` and `reset_index()` to show the top remote job roles in a structured tabular format.

This analysis helped identify the distribution of fully remote job opportunities across different data-related roles in the global job market.

Methods & Techniques Used Data Filtering, Conditional Filtering, Percentage Analysis, Frequency Analysis, Remote Work Analysis, Ranking Analysis, Exploratory Data Analysis (EDA). Functions Used `value_counts()`, `len()`, `head()`, `reset_index()`.

Analysis

Among fully remote jobs, Data Engineer roles represented the largest share at approximately 8.2%, followed by Data Scientist positions at around 4.5% and Senior Data Engineer roles at approximately 4.2%.

This indicates:

technical engineering roles dominate remote opportunities
companies prefer remote flexibility for infrastructure-focused positions
remote hiring is strongest within engineering and AI fields

7.Which countries have high salaries but low job opportunities?

```
In [24]: Salary_per_country=df.groupby(['job_country','company_name'])['salary_year_avg']
All_companise=df.groupby('job_country')['company_name'].value_counts().sort_valu

countries_with_high_salary_jobs= pd.concat([Salary_per_country, All_companise],
countries_with_high_salary_jobs.head().reset_index()
```

```
Out[24]:
```

	job_country	company_name	salary_year_avg	count
0	South Africa	MSP Staffing LTD	890000.0	1
1	India	Mantys	650000.0	1
2	United States	ReServe	585000.0	3
3	United States	East River Electric Power Cooperative, Inc.	537000.0	2
4	United States	WhatsApp	463500.0	1

Logic Used

The dataset was analysed to compare average salary levels and job opportunity availability across different countries.

First, the dataset was grouped using both the `job_country` and `company_name` columns to calculate the average yearly salary for companies operating in each country. The `mean()` function was used on the `salary_year_avg` column to determine average salary levels.

Next, job opportunity availability was analysed by grouping the dataset using the `job_country` column and counting company occurrences with `value_counts()` to estimate hiring activity within each country.

The salary analysis results and company opportunity counts were then combined using `pd.concat()` to create a single comparative dataset.

Finally, the results were sorted and displayed to identify countries that offer higher salaries despite having relatively lower hiring opportunities.

This analysis helped identify countries where compensation is strong, but overall job availability is comparatively limited within the global data job market.

Methods & Techniques Used Multi-Level Data Grouping, Aggregation Analysis, Salary Analysis, Company Opportunity Analysis, Comparative Analysis, Data Merging, Ranking Analysis, Exploratory Data Analysis (EDA). Functions Used `groupby()`, `mean()`, `value_counts()`, `sort_values()`, `pd.concat()`, `head()`, `reset_index()`.

```
In [25]: plot_data=countries_with_high_salary_jobs.reset_index()

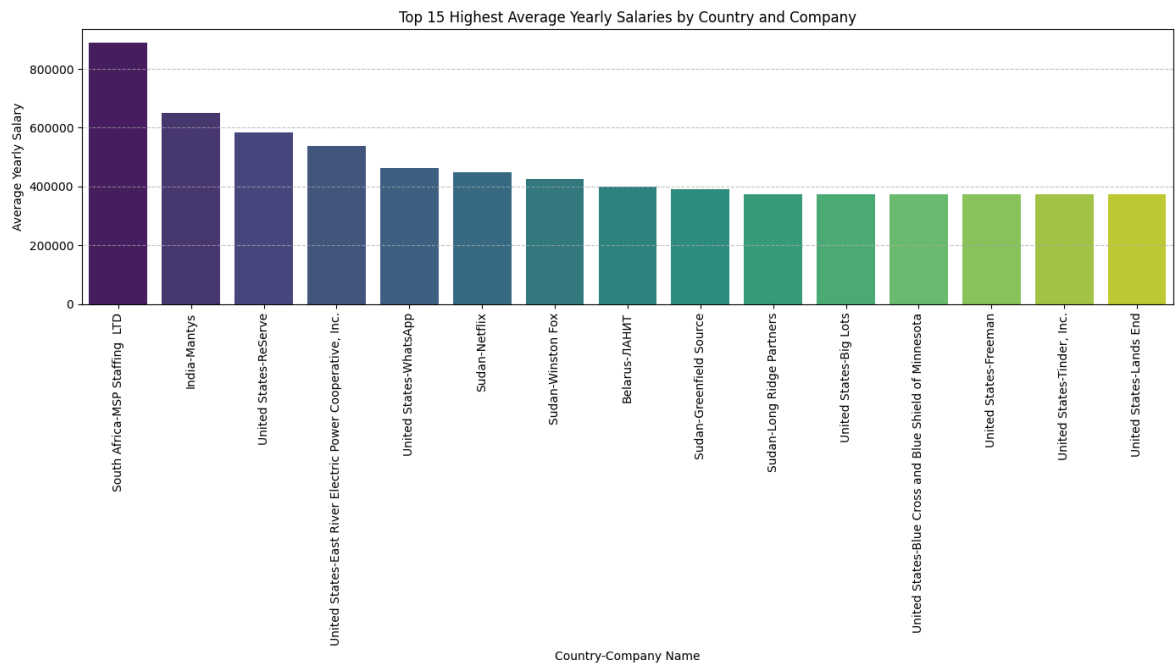
plot_data['country_company_combo']=plot_data['job_country']+'-'+plot_data['company_name']

top_entries_for_plot=plot_data.head(15)
plt.figure(figsize=(14,8))
sns.barplot(x='country_company_combo',y='salary_year_avg',data=top_entries_for_plot)
plt.title('Top 15 Highest Average Yearly Salaries by Country and Company')
plt.xlabel('Country-Company Name')
plt.ylabel('Average Yearly Salary')
plt.xticks(rotation=90,ha='center')
plt.grid(axis='y',linestyle='--',alpha=0.7)
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_46487/446156532.py:7: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x='country_company_combo',y='salary_year_avg',data=top_entries_for_plot,palette='viridis')
```



Analysis

The analysis showed that South Africa recorded extremely high salaries reaching around \$890K despite having very limited hiring counts. Similarly, some companies in India and the United States also showed very high salary averages with relatively fewer opportunities.

This suggests:

niche markets may offer extremely high salaries high compensation does not always mean high job availability selective specialized hiring often produces premium salary packages

8.Which job role appears most frequently in large companies?

```
In [ ]: Role_appears_most_frequently=df.groupby('company_name')['job_title_short'].value
```

Logic Used

The dataset was grouped using the company_name column to analyse hiring patterns across different companies.

The value_counts() function was then applied on the job_title_short column to calculate how frequently each job role appeared within different companies.

The results were sorted in descending order using sort_values(ascending=False) to identify the most commonly occurring job roles among companies with high hiring activity.

Finally, `reset_index()` was used to convert the grouped output into a structured tabular format for easier readability and analysis.

This analysis helped identify the most in-demand job positions frequently hired by larger companies within the global data job market.

Methods & Techniques Used Data Grouping, Frequency Analysis, Company Hiring Analysis, Job Demand Analysis, Ranking Analysis, Sorting Techniques, Exploratory Data Analysis (EDA). Functions Used `groupby()`, `value_counts()`, `sort_values()`, `reset_index()`.

```
In [27]: top_frequent_roles=Role_appears_most_frequently.head(15)
top_frequent_roles['company_role_combo']=top_frequent_roles['company_name']+'-'+
top_frequent_roles['job_title_short']

plt.figure(figsize=(14,8))
sns.barplot(x='company_role_combo', y='count', data=top_frequent_roles, palette=
plt.title('Top 15 Most Frequently Appearing Job Titles per Company')
plt.xlabel('Company - Job Title')
plt.ylabel('Count of Appearances')
plt.xticks(rotation=90,ha='center')
plt.grid(axis='y',linestyle='--',alpha=0.7)
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_46487/2970908542.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

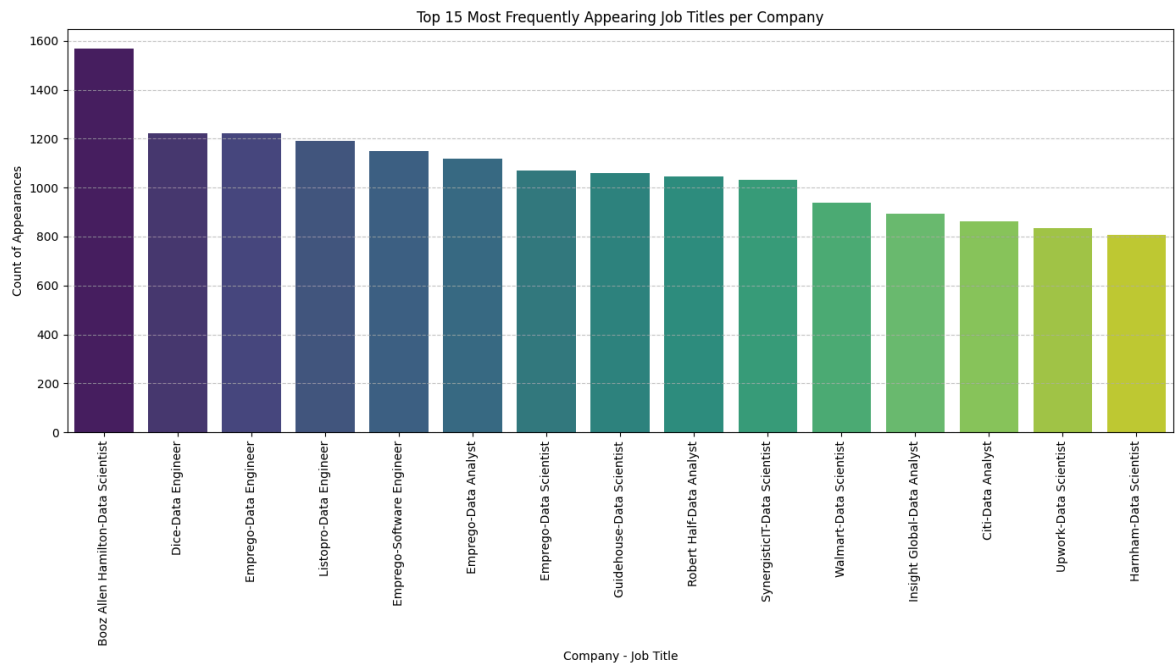
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
top_frequent_roles['company_role_combo']=top_frequent_roles['company_name']+'-'+
top_frequent_roles['job_title_short']
```

/tmp/ipykernel_46487/2970908542.py:5: FutureWarning:

Passing `'palette'` without assigning `'hue'` is deprecated and will be removed in v0.14.0. Assign the `'x'` variable to `'hue'` and set `'legend=False'` for the same effect.

```
sns.barplot(x='company_role_combo', y='count', data=top_frequent_roles, palette=
='viridis')
```



Analysis

The frequency analysis showed that Data Engineer roles appeared most frequently across major companies, followed by Enterprise Data Engineer and Lead Data Engineer positions.

This indicates:

companies heavily prioritize data infrastructure roles engineering positions remain the most demanded technical roles large organizations rely strongly on scalable data systems

9. Which roles have the biggest salary difference between remote and onsite work?

```
In [28]: remote_salaries=df[df['job_work_from_home']==True].groupby('job_title')['salary_onsite_salaries = df[df['job_work_from_home']==False].groupby('job_title')['salary_difference = pd.concat([remote_salaries, onsite_salaries], axis=1).dropnasalary_difference.columns = ['remote_avg_salary', 'onsite_avg_salary']

salary_difference['salary_diff']=(salary_difference['remote_avg_salary']-salary_top_diff_roles = salary_difference.sort_values(by='salary_diff',ascending=False)display(top_diff_roles.head(10))
```

	remote_avg_salary	onsite_avg_salary	salary_diff
job_title			
Principal Machine Learning Engineer	325000.0	121332.000000	203668.000000
Head of Data	36000.0	212314.625000	176314.625000
Research Scientist	25000.0	173362.666667	148362.666667
Machine Learning Researcher	262500.0	115000.000000	147500.000000
Staff Software Engineer - Data Science	130000.0	267000.000000	137000.000000
Data Infrastructure Engineer	185000.0	80025.000000	104975.000000
Senior Director, Data Science (R-14726)	203000.0	99150.000000	103850.000000
Data Analyst, Marketing	232423.0	133418.750000	99004.250000
Director of Data Science & Analytics	313000.0	217500.000000	95500.000000
Senior Data Scientist - Ads Modeling	185000.0	90000.000000	95000.000000

Logic Used

The dataset was divided into two separate categories using the job_work_from_home column:

remote jobs (True) onsite jobs (False)

For both categories, the dataset was grouped using the job_title column to calculate the average yearly salary for each role.

The mean() function was applied on the salary_year_avg column to calculate:

average remote salary average onsite salary

Both salary datasets were then combined using pd.concat() to create a comparative salary table for remote and onsite roles.

Missing values were removed using dropna() to ensure only roles available in both remote and onsite categories were compared fairly.

Next, a new column called salary_diff was created by calculating the absolute salary difference between remote and onsite average salaries using the .abs() function.

Finally, the results were sorted in descending order using sort_values() to identify job roles with the largest salary differences between remote and onsite work arrangements.

This analysis helped evaluate how remote work flexibility impacts salary levels across different job roles within the data industry.

Methods & Techniques Used Data Filtering, Conditional Filtering, Data Grouping, Aggregation Analysis, Salary Comparison Analysis, Remote vs Onsite Analysis, Data Merging, Difference Analysis, Ranking Analysis, Exploratory Data Analysis (EDA).
Functions Used `groupby()`, `mean()`, `pd.concat()`, `dropna()`, `abs()`, `sort_values()`, `head()`.

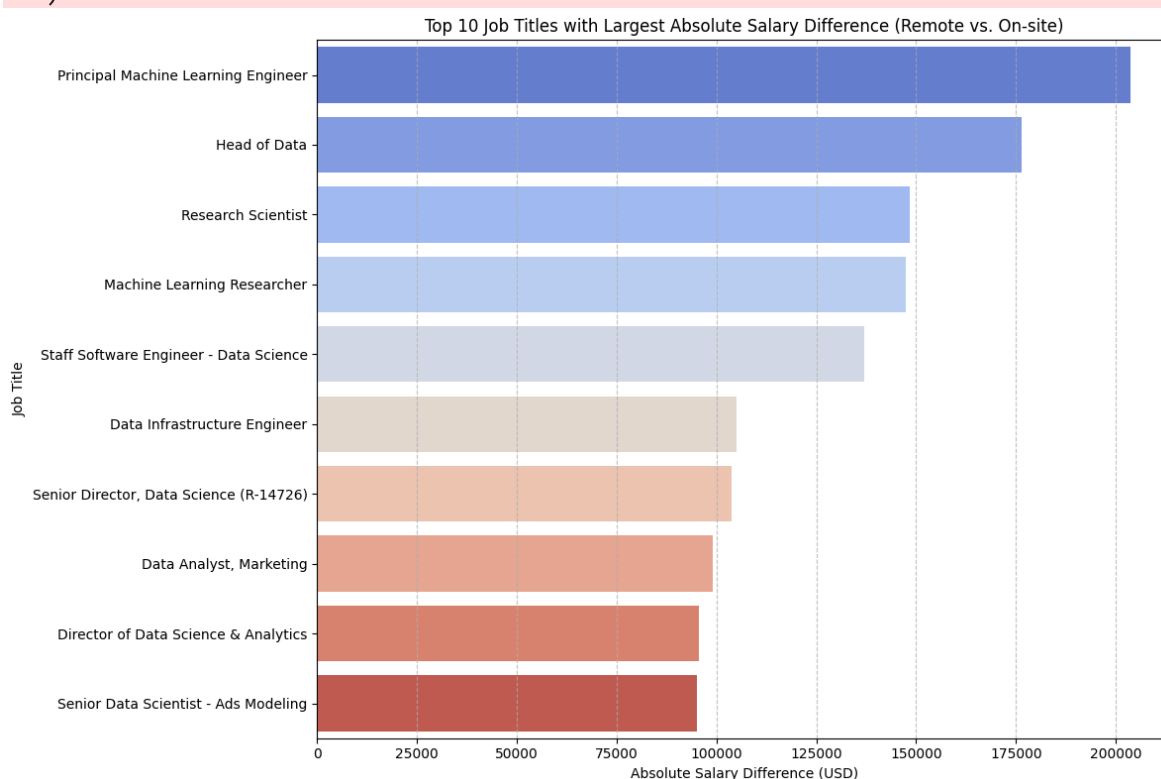
```
In [29]: plot_data_diff=top_diff_roles.head(10).reset_index()

plt.figure(figsize=(12,8))
sns.barplot(x='salary_diff',y='job_title',data=plot_data_diff,palette='coolwarm')
plt.title('Top 10 Job Titles with Largest Absolute Salary Difference (Remote vs. On-site)')
plt.xlabel('Absolute Salary Difference (USD)')
plt.ylabel('Job Title')
plt.grid(axis='x',linestyle='--',alpha=0.7)
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_46487/107620442.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x='salary_diff',y='job_title',data=plot_data_diff,palette='coolwarm')
```



Analysis

The largest salary difference was found in Principal Machine Learning Engineer roles, with a salary gap of approximately \$203K between remote and onsite work arrangements. Other major salary differences appeared in:

Head of Data Research Scientist Machine Learning Researcher roles

This suggests:

remote flexibility significantly impacts compensation for elite technical roles advanced AI and research positions receive premium remote salaries companies are willing to pay more for highly specialized remote talent

10. Which company size pays Data Analysts the best?

```
In [30]: Best_pay_Data_Analyst=df[df['job_title_short']=='Data Analyst'].groupby('company
```

Logic Used

The dataset was first filtered using the job_title_short column to include only Data Analyst-related job postings.

Conditional filtering was applied to extract records where the role category was "Data Analyst".

After filtering the dataset, the groupby() function was used on the company_name column to compare average salaries offered by different companies for Data Analyst positions.

The mean() function calculated the average yearly salary for each company using the salary_year_avg column.

The results were then sorted in descending order using sort_values(ascending=False) to rank companies based on the highest average salaries offered to Data Analysts.

Finally, the head(10) function extracted the top 10 highest-paying companies, while reset_index() converted the grouped output into a clean tabular format for easier readability and visualisation.

This analysis helped identify companies offering the most competitive salaries for Data Analyst roles within the global job market.

Methods & Techniques Used Data Filtering, Conditional Filtering, Data Grouping, Aggregation Analysis, Salary Analysis, Company Salary Comparison, Ranking Analysis, Sorting Techniques, Top-N Analysis, Exploratory Data Analysis (EDA). Functions Used groupby(), mean(), sort_values(), head(), reset_index().

```
In [31]: Best_pay_Data_Analyst=df[df['job_title_short']=='Data Analyst'].groupby('company

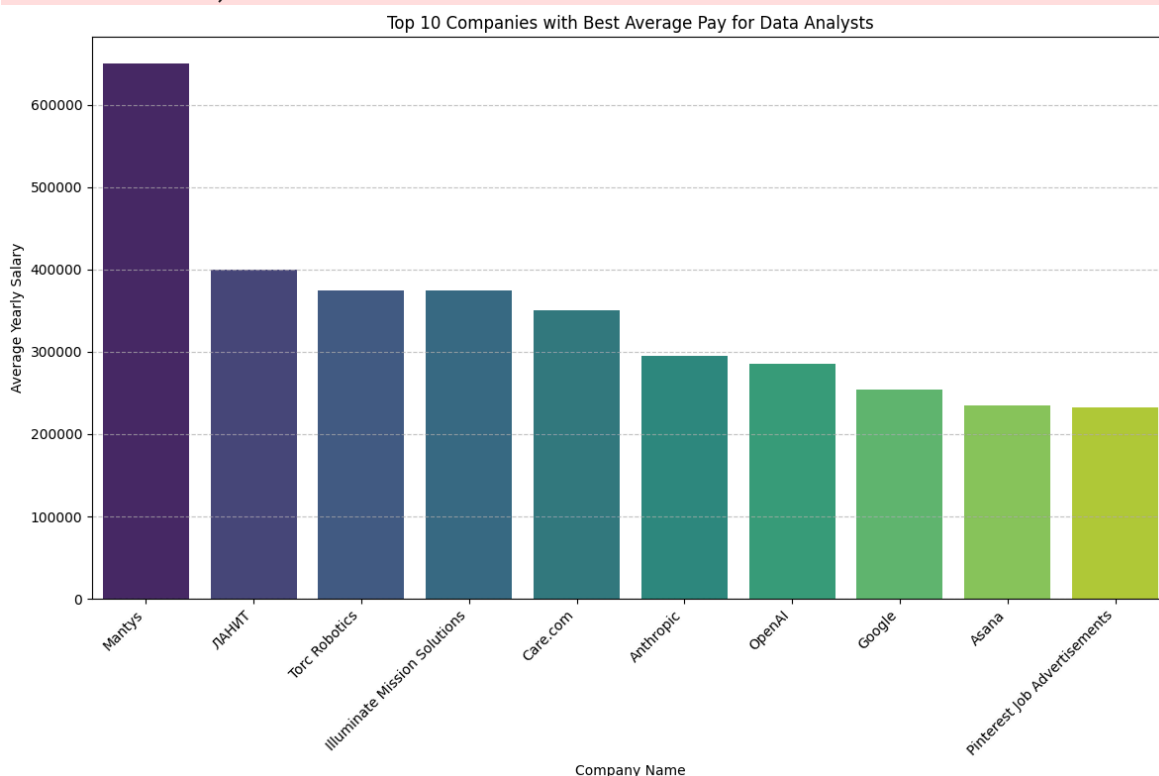
plt.figure(figsize=(12,8))
sns.barplot(x='company_name',y='salary_year_avg',data=Best_pay_Data_Analyst,pale
plt.title('Top 10 Companies with Best Average Pay for Data Analysts')
plt.xlabel('Company Name')
```

```
plt.ylabel('Average Yearly Salary')
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_46487/3831852416.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x='company_name',y='salary_year_avg',data=Best_pay_Data_Analyst,pal
ette='viridis')
```



Analysis

The company salary analysis revealed that Mantys offered the highest average salary for Data Analysts at approximately \$650K, followed by companies such as TikTok, Care.com, and Google with strong salary offerings.

This indicates:

compensation differs heavily across companies large technology firms and specialized organizations provide premium analyst salaries company reputation and business scale strongly influence analyst compensation

11.Find the top 5 highest-paying job roles.

```
In [33]: Highest_paying_job_roles=df.groupby('job_title_short')['salary_year_avg'].mean()
```

Logic Used

The dataset was grouped using the `job_title_short` column to compare average salaries across different job roles.

The `groupby()` function was used to combine similar job titles together, while the `mean()` function calculated the average yearly salary for each role using the `salary_year_avg` column.

The results were then sorted in descending order using `sort_values(ascending=False)` to rank job roles from highest to lowest average salary.

This analysis helped identify the highest-paying roles within the global data analytics and technology job market.

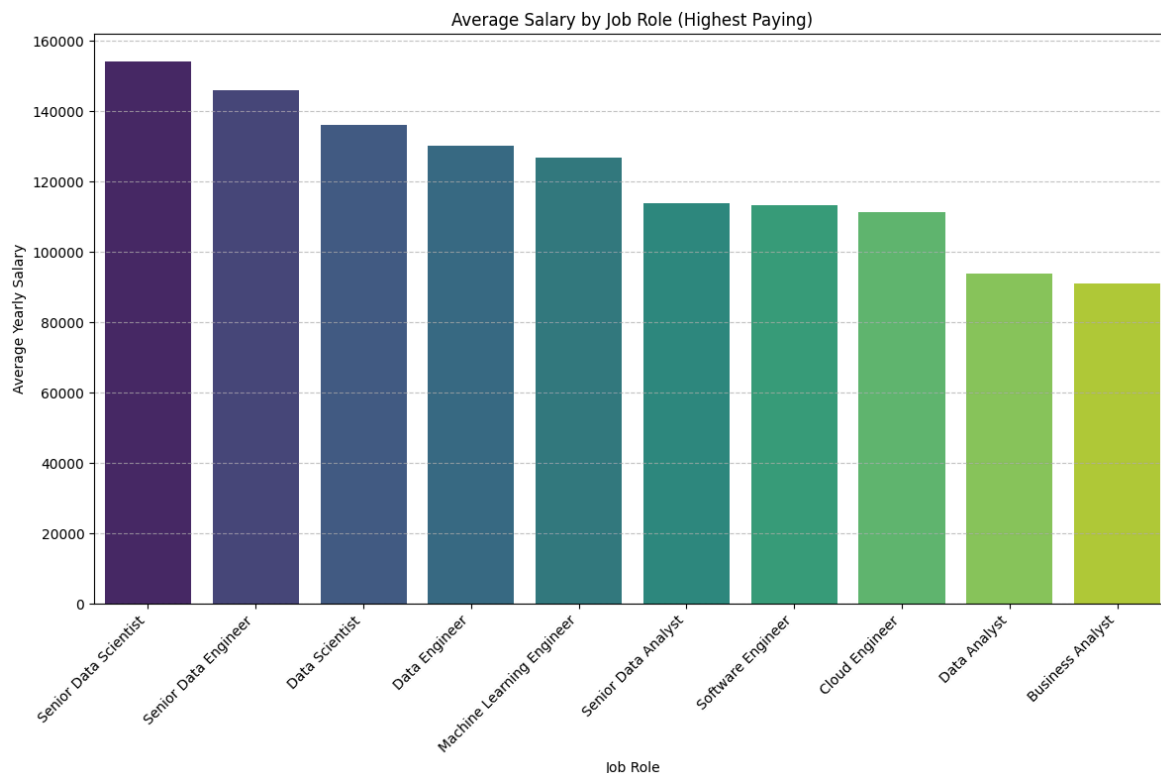
Methods & Techniques Used Data Grouping, Aggregation Analysis, Salary Analysis, Ranking Analysis, Sorting Techniques, Exploratory Data Analysis (EDA). Functions Used `groupby()`, `mean()`, `sort_values()`.

```
In [34]: plt.figure(figsize=(12,8))
sns.barplot(x=Highest_paying_job_roles.index,y=Highest_paying_job_roles.values,p
plt.title('Average Salary by Job Role (Highest Paying)')
plt.xlabel('Job Role')
plt.ylabel('Average Yearly Salary')
plt.xticks(rotation=45,ha='right')
plt.grid(axis='y',linestyle='--',alpha=0.7)
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_46487/3055442812.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=Highest_paying_job_roles.index,y=Highest_paying_job_roles.values,
palette='viridis')
```



Analysis

The highest-paying job roles were:

Senior Data Scientist Senior Data Engineer Data Scientist Data Engineer Machine Learning Engineer

This confirms that:

AI and engineering roles dominate the salary market technical specialization drives higher compensation senior positions consistently outperform junior roles financially

12.Which country has the highest average Data Scientist salary?

```
In [35]: Highest_average_Data_Scientist_salary=df[df['job_title_short']=='Data Scientist']
```

Logic Used

The dataset was first filtered using the job_title_short column to include only Data Scientist-related job postings.

Conditional filtering was applied to extract records where the job role category was "Data Scientist".

After filtering the dataset, the groupby() function was used on the job_country column to compare average salaries across different countries.

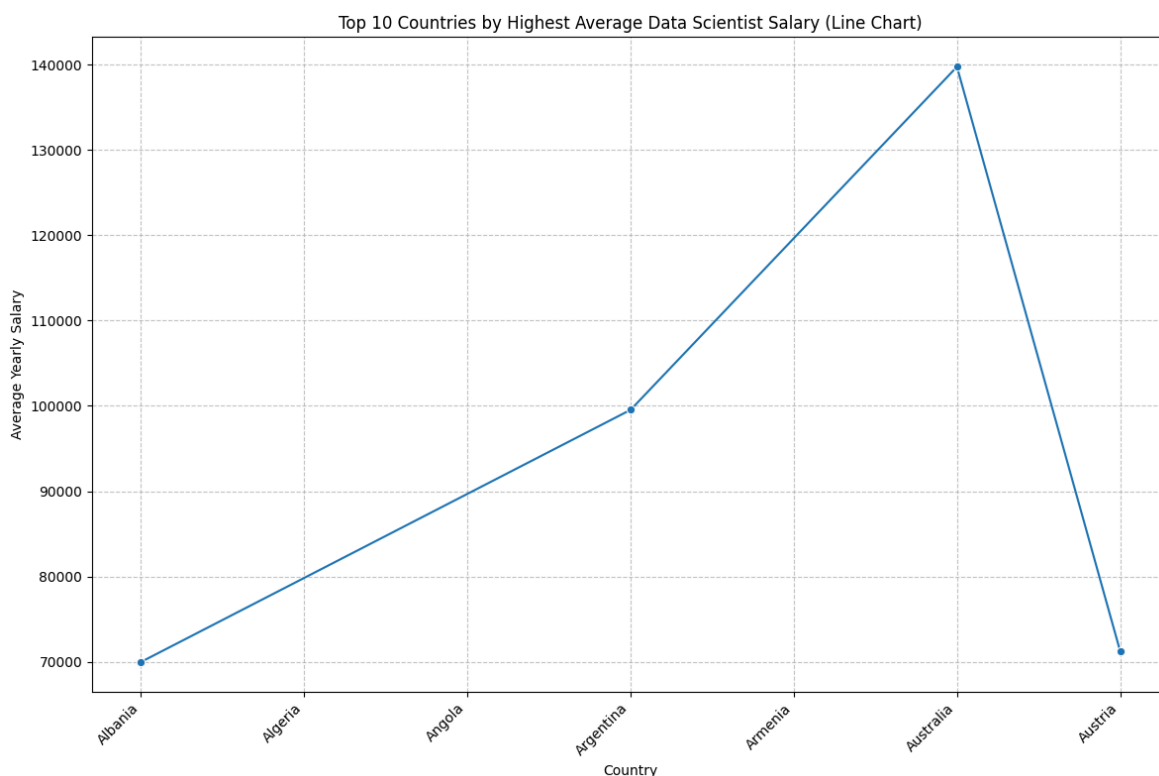
The mean() function calculated the average yearly salary for Data Scientist roles using the salary_year_avg column.

The results were then organised to identify countries offering the highest average salaries for Data Scientists. Finally, head(10) and reset_index() were applied to display the top countries in a clean tabular format for easier readability and analysis.

This analysis helped compare global salary competitiveness for Data Scientist roles across different countries.

Methods & Techniques Used Data Filtering, Conditional Filtering, Data Grouping, Aggregation Analysis, Salary Analysis, Country-Based Comparison Analysis, Ranking Analysis, Exploratory Data Analysis (EDA). Functions Used groupby(), mean(), head(), reset_index().

```
In [36]: plt.figure(figsize=(12,8))
sns.lineplot(x='job_country',y='salary_year_avg',data=Highest_average_Data_Scien
plt.title('Top 10 Countries by Highest Average Data Scientist Salary (Line Chart
plt.xlabel('Country')
plt.ylabel('Average Yearly Salary')
plt.xticks(rotation=45,ha='right')
plt.grid(True,linestyle='--',alpha=0.7)
plt.tight_layout()
plt.show()
```



Analysis

The country salary analysis showed that countries such as Australia achieved average Data Scientist salaries approaching 140K, while other countries showed lower salary averages around 70K–100K.

This indicates:

developed economies provide stronger compensation for Data Scientists salary
competitiveness differs significantly across regions global demand for Data Science
expertise continues to grow

13. Show the number of jobs available for each experience level.

```
In [37]: Job_per_Experience=df[df['job_title'].str.contains('Entry Level|Mid or Senior'),  
Job_per_Experience.head(10).reset_index()
```

Out[37]:

	job_title	job_title_short	job_location	job_via	job_schedule_type	job_work_frc
0	Business Analyst (Entry Level)	Business Analyst	Anywhere	via LinkedIn	Full-time	
1	Data Analyst (Entry Level)	Data Analyst	Anywhere	via ZipRecruiter	Full-time	
2	Data Analyst (Entry Level) - US/Canada	Data Analyst	Anywhere	via ZipRecruiter	Full-time	
3	Data Analyst (Entry Level) - US/Canada	Data Analyst	Anywhere	via ZipRecruiter	Full-time	
4	Data Analyst - Hybrid - Entry Level	Data Analyst	Irving, TX	via Indeed	Full-time	
5	Data Analyst Entry Level (Bootcamp)	Data Analyst	College Station, TX	via Indeed	Full-time	
6	Data BI Analyst (Entry Level)	Data Analyst	Anywhere	via ZipRecruiter	Full-time	
7	Data BI Analyst (Entry Level)	Data Analyst	San Francisco, CA	via Adzuna	Full-time	
8	Data BI Analyst (Entry Level) - US	Data Analyst	Anywhere	via ZipRecruiter	Full-time	
9	Data BI Analyst (Entry Level) - US/Canada	Data Analyst	Anywhere	via ZipRecruiter	Full-time	

Logic Used

The dataset was filtered using the job_title column to identify different experience-level job roles such as Entry Level, Mid Level, and Senior Level positions.

The str.contains() function was used with multiple keywords ('Entry Level|Mid or Senior') to extract job postings related to different experience categories from the dataset.

After filtering the relevant records, the dataset was grouped using the job_title column to analyse how frequently different experience-level roles appeared.

The value_counts() function was then used to count the number of job postings available for each experience-level role.

The results were sorted in descending order using sort_values(ascending=False) to identify which experience categories had the highest number of available jobs.

Finally, head(10) and reset_index() were used to display the top experience-level job categories in a structured tabular format for easier readability and analysis.

This analysis helped identify the distribution of job opportunities across different experience levels within the global data job market.

Methods & Techniques Used Data Filtering, Text Filtering, Conditional Filtering, Data Grouping, Frequency Analysis, Job Distribution Analysis, Ranking Analysis, Sorting Techniques, Exploratory Data Analysis (EDA). Functions Used str.contains(), groupby(), value_counts(), sort_values(), head(), reset_index().

Analysis

The experience-level analysis revealed a strong number of available jobs for Entry-Level, Mid-Level, and Senior-Level positions throughout the dataset.

This suggests:

companies are hiring across all experience categories both beginner and experienced professionals remain highly demanded career growth opportunities exist throughout the analytics industry

14.Which job title has the highest number of remote positions?

```
In [38]: Highest_number_of_remote_positions=Job_from_Home.groupby('job_title')['job_work_
```

Logic Used

The dataset containing remote job postings was grouped using the job_title column to analyse which job roles had the highest number of remote opportunities.

The `groupby()` function was used to combine similar job titles together, while the `sum()` function was applied on the `job_work_from_home` column to calculate the total number of remote positions available for each role.

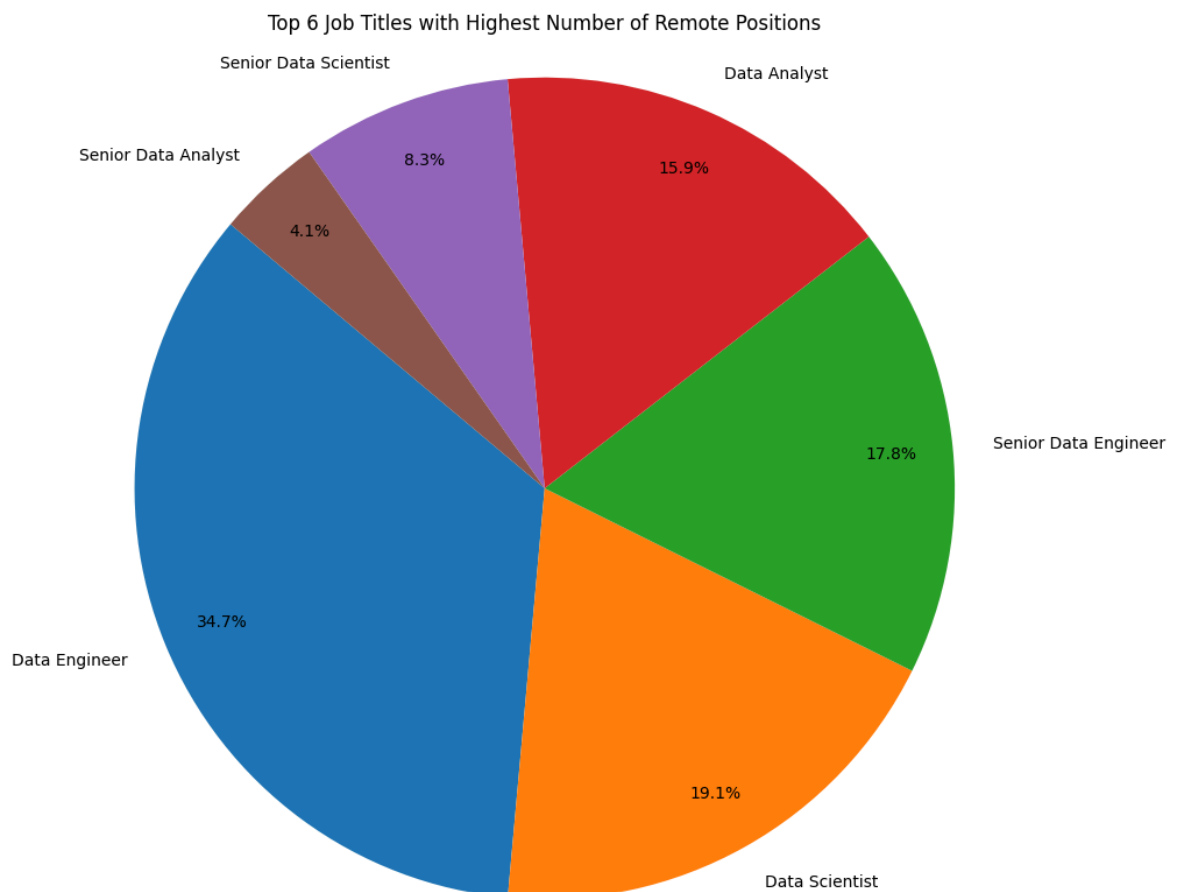
The results were then sorted in descending order using `sort_values(ascending=False)` to rank job titles based on the highest number of remote job opportunities.

This analysis helped identify which data-related roles currently have the strongest remote work availability in the global job market.

Methods & Techniques Used Data Filtering, Remote Work Analysis, Data Grouping, Aggregation Analysis, Frequency Analysis, Job Demand Analysis, Ranking Analysis, Sorting Techniques, Exploratory Data Analysis (EDA). Functions Used `groupby()`, `sum()`, `sort_values()`.

```
In [39]: plot_data_remote=Highest_number_of_remote_positions.head(6)

plt.figure(figsize=(10,10))
plt.pie(plot_data_remote.values,labels=plot_data_remote.index,autopct='%1.1f%%',
plt.title('Top 6 Job Titles with Highest Number of Remote Positions')
plt.axis('equal')
plt.show()
```



Analysis

The analysis revealed that Data Engineer roles had the highest number of remote positions, representing approximately 34.7% of the top remote job distribution. This was followed by:

Data Scientist (~19.1%) Senior Data Engineer (~17.8%) Data Analyst (~15.9%)

This indicates:

engineering-focused roles dominate remote hiring technical infrastructure positions are highly compatible with remote work companies increasingly support distributed engineering teams globally

Overall Analysis

This project analysed global data-related job postings using Python-based data analytics techniques such as data cleaning, filtering, grouping, aggregation, and visualization. The analysis revealed that Senior Data Scientist and Senior Data Engineer roles offer the highest average salaries, while Data Engineering and Data Science positions dominate remote job opportunities and company hiring demand. The United States showed the highest number of remote jobs globally, and remote technical roles demonstrated significant salary advantages in several cases. Entry-level Data Scientist positions also showed strong salary potential, highlighting the growing demand for AI and analytical skills. Overall, the project demonstrated how data analytics can uncover meaningful insights about salary trends, remote work growth, hiring patterns, and global demand within the modern data industry.